

Same Form, Different Grammars: The Slavic [ov]~[u] Alternation

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Abstract

This paper investigates the [ov]~[u] alternation in Slavic verbs derived from nominal and adjectival stems. In addition to the alternation between a sequence and a single segment, Slavic languages differ with respect to the surface shape of the VC sequence: some only feature [ov], some alternate [ov] and [ev] depending on the preceding consonant, some only feature [uv]. Drawing from both synchronic and diachronic predecessors, it is shown that the alternation deserves a unified approach. The verbaliser suffix has an abstract underlying representation, adapted phonologically based on hiatus resolution requirements: an underlyingly defective diphthong surfaces as a vowel before a consonant and as a sequence before a vowel. Inter-Slavic variation is ascribed to nuanced preferences in the phonological grammar, which provides an insight into the shared properties within the Slavic linguistic family, like the markedness of [±back] features.

Keywords: denominal verbs, Slavic, diphthong, hiatus resolution, variation

1 Introduction

Verbs in Slavic derived from nominal and adjectival bases exhibit the [ov]~[u] alternation of the VBLZ suffix depending on the verbal base (1). In the infinitival stem (henceforth: INF), employed for forms such as the active past participle or the so-called *l*-form, these verbs are always found with a VC sequence (1)-a, but in the present tense stem (henceforth: PRS), a single vowel is found instead. In Bosnian/Croatian/Montenegrin/Serbian, here taken as a representative of the South Slavic branch, this suffix is additionally altered to conform to the quality of the preceding palatal consonant (1)-b, which does not affect the quality of the single vowel employed in the PRS stem.

(1) *Bosnian/Croatian/Montenegrin/Serbian denominal verbs*

	L-FORM.F.SG	2SG.PRS	gloss
a.	[neg- ov -a-l-a]	[neg- u -je-ʃ]	‘care for’
	[dar- ov -a-l-a]	[dar- u -je-ʃ]	‘gift’
b.	[matʃ- ev -a-l-a]	[matʃ- u -je-ʃ]	‘fence’
	[tsar- ev -a-l-a]	[tsar- u -je-ʃ]	‘reign’

The alternation illustrated in (1) is widespread in Slavic, with some languages featuring only [ov], some alternating between [ov] and [ev] like BCMS, some featuring only [uv]. The present tense stem always features [u], regardless of the VC sequence. This indicates that there is a deep relationship between the Slavic languages when it comes to this alternation.

Numerous diachronical studies have traced the origin of the alternation in (1) to Common Slavic, Balto-Slavic, even Proto-Indo-European (PIE). The diachronic studies meet in one point crucial for the proposal of this paper: they agree that all realisations of the VBLZ have a common historical origin in a complex item, a sort of diphthong or an overspecified segment, adapted to different surface forms to satisfy phonological restrictions on segmental co-occurrence.

From the synchronic perspective, the [ov]~[u] alternation was previously ascribed to iotation (Rubach 1984), listed allomorphs (Halle 1963; Rubach and Booij 2001), phonotactics (Marković 2013), morpheme-specific assimilation rules (Coats 1974), the development of *yer* vowels (Matushansky 2009; Pesetsky 1979), complex vowel sequences (Melvold 1989), and, most recently, floating segments (Simonović et al. 2023). The synchronic studies primarily concern individual languages and do not express aims to provide a unified solution for a widespread Slavic alternation. The second disadvantage of these approaches is that the analyses proposed for one type of VBLZ suffixes are usually not extendable to other types (apart from the approaches based on selection of listed allomorphs).

This paper proposes taking a path similar to the diachronic studies. Namely, it is emphasised in the diachronic studies that a pan-Slavic alternation has a unified genesis, therefore a unified analysis is needed. For this reason it is proposed here that the alternation in focus has a dual origin: representational and computational. Inspired by Inkelas et al.’s (1997) *Prespecification Method*, Feldstein’s (2003) work on Slavic monophthongisation processes, and Harris’s (1985) work on Spanish diphthongs, the present analysis suggests that the VBLZ is a single morphological unit with a complex phonological exponent.

Specifically, I argue that the the [ov]~[u] alternation originates in an abstract underlying representation (henceforth: UR) of the VBLZ suffix, stored and available for all Slavic languages. The VBLZ suffix is underlyingly represented as an incomplete diphthong, which is adapted to the phonological environment according to hiatus resolution requirements, such that the single vowel appears before a consonant and the VC sequence surfaces before a vowel. The differences between the three groups of Slavic languages based on the surface shape of the VC sequences are here attributed to the computational properties of their phonological grammars, formalised as three minimally different Optimality-theoretic systems (Prince and Smolensky 1993; Zubritskaya 1995). The inter-Slavic variation is thus shown to arise under group-specific preferences for giving different levels of priority to autosegmentally represented phonological operations of feature copying (modelled as sharing) and feature epenthesis.

This paper is structured as follows. The [ov]~[u] alternation patterns in several Slavic languages are presented in section 2, showing that the alternation is not morphologically, but rather phonologically conditioned. The foundation for the present analysis is built in section 3, by discussing the advantages and disadvantages of previous attempts to investigate the [ov]~[u] alternation from both the diachronic and the synchronic perspective. Section 4 introduces the analytical idea of the paper, that the observed alternation, previously seen as allomorphy, should be attributed to phonological derivation the abstract UR of the VBLZ suffix is subject to. Subsections (4.1)–(4.4) provide details on phonological derivations of contexts in which the said abstract UR of the VBLZ is found: the pre-consonantal [u], the [ov]-group, the [ov]/[ev] group, the [uv] group. A discussion of the proposed grammars is given in subsection 4.5, while section 5 summarises the analysis and concludes the paper.

2 Deadjectival and denominal verbs across Slavic

Verbs derived from nominal and adjectival bases using this verbalizing suffix¹ show distinct infinitive (INF) and present tense (PRS) stem forms. The INF stem, used in participles, has a VC sequence, while the PRS stem has a single vowel. This alternation, rooted in Proto-Slavic (Schuyt 1990: 395), also appears in modern Baltic languages, as in Lithuanian *-áuti/-áuja* (Ambrazas et al. 2006) and Latvian *-aut/-auju/-avu* verbs (Praulīš 2012).

Table 1 provides an overview of denominal/deadjectival verbs in two indicative forms: the *l*-form, corresponding to the Proto-Slavic *l*-participle used for past tenses, based on the INF stem, and the second person singular for present tense stem. The feminine *l*-form is used to avoid potential confusion from the fate of word-final /-l/ for masculine. As part of derivational morphology, the VBLZ suffix appears between the nominal/adjectival base and inflectional morphology. The verbs display a vowel-consonant sequence in the *l*-form but a single vowel [u] in the present tense.

language	L-FORM.F.SG	2SG.PRS	gloss
Belarusian	[kas-<av>-a-l-a]	[kas-u-je-ʃ]	‘abolish’
Bos/Cro/Mon/Ser	[neg-ov-a-l-a]	[neg-u-je-ʃ]	‘care for’
Bulgarian	[tsel-uv-a-l-a]	[tsel-uv-a-ʃ]	‘kiss’
Bunjev(ac)	[bog-ov-a-l-a]	[bog-u-je-ʃ]	‘swear at god’
Carpathian Rusyn	[bisid-ov-a-l-a]	[bisid-u-je-ʃ]	‘talk’
Czech	[kup-ov-a-l-a]	[kup-u-je-ʃ]	‘buy’
Kashubian	[brutk-ov-a-w-a]	[brutk-u-je-ʃ]	‘get engaged’
Lemko Rusyn	[ris-uv-a-l-a]	[ris-u-je-ʃ]	‘draw’
Lower Sorbian	[mol-ov-a-w-a]	[mol-u-je-ʃ]	‘paint’
Macedonian	[ver-uv-a-l-a]	[ver-uv-a-ʃ]	‘believe’
Pannonian Rusyn	[rab-ov-a-l-a]	[rab-u-je-ʃ]	‘serve’
Podlachian	[tsil-ov-a-w-a]	[tsil-u-je-ʃ]	‘aim’
Polish	[prats-ov-a-w-a]	[prats-u-je-ʃ]	‘work’
Russian	[ʒal-<ov>-a-l-a-sʲ]	[ʒal-u-je-ʃ-sʲ]	‘complain’
Silesian	[ɕtop-ov-a-w-a]	[ɕtop-u-je-ʃ]	‘darn’
Slovak	[boj-ov-a-l-a]	[boj-u-je-ʃ]	‘battle’
Slovenian	[pot-ov-a-l-a]	[pot-u-je-ʃ]	‘travel’
Ukrainian	[stis-uv-a-l-a]	[stis-u-je-ʃ]	‘squeeze’
Upper Sorbian	[kup-ov-a-w-a]	[kup-u-je-ʃ]	‘buy’

Table 1: Overview of [Vv]~[u] verbs in Slavic

The overview in Table 1 shows that the alternation is fairly consistent across the examined languages. The VC sequence [av]/[ov]/[uv] is observed in the INF stem, followed by /-a-/ and the inflection for F.SG of the *l*-form. In the PRS stem, [u] is predominantly used in lieu of the VC sequence, now preceding the theme vowel /-je-/. Only Bulgarian and Macedonian deviate in the present tense by consistently featuring the VC sequence followed by /-a-/.²

The form <av> in Belarusian likely originates from /o/, surfacing as [a] due to vowel reduction (Crosswhite 2001; Goodman Finedore and Scatton 1978; Iosad 2012; Mołczanow 2007; Nesset 2002, inter alia), unique to Belarusian orthography. Reports indicate that vowel reduction in Belarusian and Russian, (Southern Great Russian dialects) typically results in [ə] (or [i]

¹This suffix is one of many in Slavic with similar phonological behaviour. In Macedonian, Polish, Ukrainian, it also functions for secondary imperfectives. The morpho-syntactic nature is not much related to the alternations the suffix exhibits, given the division of grammatical domains assumed here (Bermúdez-Otero 2012).

²Although see Velkova (2011) for a report on an optional contraction of [uv] to [uu] in the present tense in the dialects of southwestern Bulgaria. Schuyt (1990) mentions [ov]-verbs as loanwords from Russian in Bulgarian.

after a palatal), suggesting the allomorph [əv]. This paper focuses on the form reflected in Russian orthography, used here to infer pre-reduction quality. For clarity, Belarusian and Russian forms are presented orthographically in angle brackets. Other languages in Table 1 show [ov] in the INF stem.

In some of the examined Slavic systems, the VBLZ suffix shows sensitivity to the quality of the root-final consonant. This property is also inherited from Proto-Slavic (Olander 2015). Consequently, the VBLZ suffix surfaces as [ev] in the INF stem of some Slavic languages, while still alternating with [u] in the PRS stem. As illustrated with the palatal-final bases in Table 2, four out of the seventeen examined systems feature [ev] following a palatal, whereas others maintain [ov] or [uv] regardless of the quality of the preceding consonant.

language	INF	2SG.PRS	gloss
Belarusian	[vaj-<av>-a-l-a]	[vaj- u -je-ʃ]	‘battle’
Bunjev(ac)	[katʃ- ov -a-l-a]	[katʃ- u -je-ʃ]	‘prune’
Czech	[bitʃ- ov -a-l-a]	[bitʃ- u -je-ʃ]	‘whip’
Kashubian	[vrəj- ov -a-w-a]	[vrəj- u -je-ʃ]	‘date’
Lower Sorbian	[puç- ov -a-w-a]	[puç- u -je-ʃ]	‘wander’
Pannonian Rusyn	[kraʃ- ov -a-l-a]	[kraʃ- u -je-ʃ]	‘reign’
Podlachian	[kartʃ- ov -a-w-a]	[kartʃ- u -je-ʃ]	‘grub’
Polish	[hotel’- ov -a-w-a]	[hotel’- u -je-ʃ]	‘accommodate’
Slovak	[boj- ov -a-l-a]	[boj- u -je-ʃ]	‘battle’
Silesian	[pajtɕ- ov -a-w-a]	[pajtɕ- u -je-ʃ]	‘whip’
Upper Sorbian	[preloz- ov -a-w-a]	[preloz- u -je-ʃ]	‘translate’
Bos/Cro/Mon/Ser	[uʦʲiteʃ- ev -a-l-a]	[uʦʲiteʃ- u -je-ʃ]	‘be teacher’
Carpathian Rusyn	[stupŋ- ev -a-l-a]	[stupŋ- u -je-ʃ]	‘compare’
Russian	[muxlʲ-<ev>-a-l-a]	[muxlʲ- u -je-ʃ]	‘swindle’
Slovenian	[deʒ- ev -a-l-a]	[deʒ- u -je-ʃ]	‘rain’
Bulgarian	[najeʒ- uv -a-l-a]	[najeʒ- uv -a-ʃ]	‘jitter’
Lemko Rusyn	[oboʒ- uv -a-l-a]	[oboʒ- u -je-ʃ]	‘adore’
Macedonian	[noc- uv -a-l-a]	[noc- uv -a-ʃ]	‘spend night’
Ukrainian	[zapizŋ- uv -a-l-a]	[zapizŋ- u -je-ʃ]	‘be late’

Table 2: Denominal/deadjectival verbs with palatal-final roots

Verbs with [ev] are not entirely absent from [ov]-only or [uv]-only Slavic languages; however, this [ev] appears to be a different morpheme. In [ov]-only and [uv]-only languages, verbs with [ev] do not alternate with [u] (Schuyt 1990). Non-alternating [ev] verbs also occur in BCMS, e.g., [iz-voʃ-**ev**-a-ti] ‘to be finicky’ vs. [iz-voʃ-**ev**-a-ʃ] ‘to be finicky.2SG’, with a long [e:] stressed in the INF stem. Similar prosodic distinctions are reported in contemporary Slovenian (Simonović and Mišmaš 2023), Kashubian (Makúřô 2016), and Russian (Revithiadou 1999).

Two principal generalisations can be made about the observed pattern. First, the VC sequence is found exclusively in the INF stem for the majority of Slavic languages, but for Bulgarian and Macedonian also in the PRS stem. Unless there is an independent explanation related to the loss of infinitives in these languages altogether, the morphological conditioning must be abandoned. The alternation can be described as ‘morphologically correlated’, at best. Phonologically speaking, the VC sequence is only found, regardless of the stem, when the VBLZ is followed by a vowel. When followed by the consonant-initial theme vowel /-je-/,³ the result is

³An anonymous reviewer points out that the problem disappears if [uje] is seen as an exponent of a verbaliser head. This paper follows the arguments of Matushansky (2024); Melvold (1989); Quaglia et al. (2022) that /-je-/ is a separate morpheme in the system of theme vowels and Simonović et al. (2023) in treating [u] like the alternate surface form of the suffix [ov]/[ev]/[uv].

a single vowel. Such a set of conditions points more towards a phonological derivation instead of listed allomorphy.

Thus, the VBLZ suffix alternates between [ev], [ov], and [uv] in the INF stem because it is followed by a vowel and [u] in the PRS stem because it is followed by a consonant. The vocalic reflex aside, three groups emerge from the reflexes of the VC sequence in Table 2: (i) the [ov] group, where [ov] never assimilates to the preceding palatal, (ii) the [ov]/[ev] group, where [ov] assimilates to the preceding palatal, and (iii) the [uv] group, where [uv] never assimilates to the preceding palatal.

Table 3 categorises Slavic languages based on the quality of their VC sequences into three groups, with Russian and Belarusian marked with an asterisk. The [ov]~[ev] alternation in modern Russian functions more as an orthographic rule, given active mid vowel reduction in both Belarusian and Russian (notably in dialects with *akan'e* and *jakan'e*), which blurs the alternation. However, Russian orthographic forms can hint at pre-reduction forms, relevant to this analysis, and thus Russian is provisionally included in the [ov]/[ev] group, with angle brackets in the data indicating reliance on orthographic forms. Vowel reduction is reflected directly in the Belarusian orthography, and consultations with native speakers have not shown sensitivity to preceding palatals. Consequently, Belarusian is provisionally placed in the [ov]-only group.⁴ The pluricentricity of Rusyn is reflected in the verb forms, as it spans all three observed groups (Carpathian Rusyn covers Prešov Rusyn in Slovakia and Subcarpathian Rusyn in Ukraine, the latter not recognised as a minority language, cf. Kushko (2007)). The language of the Bunyevs in northern Vojvodina is also administratively disputed (Belić 2014).

group	language	descriptive source(s)
[ov]	Belarusian*	Byrila and Šuba (1985)
	Bunjev(ac)	Bošnjaković and Sikimić (2013); Peić et al. (2018)
	Czech	Naughton (2005); Tahal (2010)
	Kashubian	Makúróť (2016)
	Lower Sorbian	Janaš (1976)
	Pannonian Rusyn	Medeši et al. (2010); Ramač et al. (1997)
	Podlachian	Maksymiuk (2012)
	Polish	Bielec (2012); Sadowska (2012)
	Silesian	Jaroszewicz (2021); Wencel (2023)
	Slovak	Mistrík (1988)
	Upper Sorbian	Schuster-Šewc (1996)
[ov]/[ev]	Bos/Cro/Mon/Ser	Jahić et al. (2000); Marković (2013)
	Carpathian Rusyn	Pugh (2009)
	Russian*	Timberlake (2004); Wade et al. (2020)
	Slovenian	Herrity (2000)
[uv]	Bulgarian	Radeva (1979); Scatton (1984)
	Lemko Rusyn	Fontański and Chomiak (2000)
	Macedonian	Koneski (1967); Mišeska Tomić (2012)
	Ukrainian	Pugh and Press (1999)

Table 3: The groups according to the VC sequence

The data for this paper are primarily drawn from reference grammars, dictionaries, and comparable descriptive sources to establish a foundational understanding of the pattern. Subsequently, national corpora are examined to identify examples and potential counterexamples to the generalisations drawn from the descriptive sources. The findings from national corpora

⁴An acoustic study would be inconclusive, as detecting frontness on a vowel following a root-final palatal may be due to phonetic adjustments, not phonological alternation.

are then compared with those from web-based corpora and social networks and, where possible, verified by native speakers.

Although some previous authors point towards phonologically conditioned allomorph selection, the subsequent sections of this paper demonstrate that the alternation of the IPFV suffix is exclusively phonologically derived. Both the VC sequence and the single vowel stem from a unique underlying representation. This representation is presumed to be universally accessible to all Slavic languages and actively transformed into the observed surface forms by the native speakers.

3 Previous approaches

The phenomenon of the [ov]~[u] alternation has gathered considerable attention in diachronic phonological studies, surpassing the attention from synchronic approaches. While the present analysis aims to present a synchronic approach, insights from diachronic perspectives on the origin and the development of this alternation are essential for comprehending the underlying processes that generate the alternation as we know it. Moreover, subsequent sections of the paper show that mechanisms appearing diachronic at first glance remain operative in contemporary Slavic languages.

The [ov]~[u] alternation likely dates back to Common Balto-Slavic, seen in Baltic and Slavic languages. Explanations differ however regarding its origin, both in denominal verbs (presumed to be primary) and secondary imperfectives (presumed to originate from denominal verbs). Jakobson (1948: 125) describes the alternation as follows: ‘before dropped /a/, the group /ov/ is regularly replaced by /uj/’.⁵ This explanation could be seen as a version of phonologically conditioned allomorph selection. Jakobson’s idea is developed in Pugh (1994: 16-17) as a ‘morphophonemic’ alternation, where the first learned form of the VBLZ is the present tense [uj], replaced with [ov] before an /-a-/, ‘truncated before vowels, and kept before consonants’.

Alternative approaches focus more on the phonological basis of /v/ in Slavic suffixes. Schmalstieg (1956) suggests /v/ arose as a glide by adding *-ati* to /u/-final stems, influencing imperfectivisation elsewhere in Slavic, thus making an analogical connection between the [ov]/[ev]/[uv] suffix and the secondary imperfectives. On the other side, Martinet (1953) rejects phonotactic glide insertion, proposing a complex suffix **-eH^w*.

The PIE origin is further supported by Stang (1942) and Vaillant (1966), who argue for a development from **-ēu-je-* through processes like **-eu-a- > *-avā- > -ova-*. Most recently, Villanueva Svensson (2014) traces the suffix to the PIE base **h₂eu_h₁-*, evolving into a denominal suffix ‘either through univerbation or, more likely, through resegmentation of denominatives from compounds with a base noun of **h₂eu_h₁-* as the second member’ (2014: 261).

Diachronic insights thus predominantly agree for a diphthongal origin of the suffix forms, where a high rounded element surfaces as a consonant before vowel-initial suffixes and a vowel before consonant-initial. Such patterns align with typical Slavic phonological developments for diphthongs of the type $\check{V}S$ (V = vowel, S = sonorant), as well as sequence-specific repairs (deletion, resyllabification, metathesis) on the road towards monophthongisation (Feldstein 2003), crucial for this paper’s proposal.

Unlike diachronic studies, which view the [ov]~[u] alternation across Slavic languages as a pan-Slavic phonological process, synchronic studies focus on individual languages and generally avoid modelling variation like [ov]~[ov]/[ev]~[uv]. While diachronic approaches consistently link surface forms to active phonological processes, synchronic analyses vary based on computational assumptions.

⁵Bear in mind that Jakobson (1948) assumes a system where the present tense is built on top of the infinitive, in which case some segments belonging to the INF are lost or dropped.

Rubach (1984) interprets the Polish alternation as evidence of a widespread iotation rule, where /j/-Insertion, /w/-Deletion, and assimilation produce forms like *głos + uj + ō*. In contrast, Rubach and Booij (2001) argue for an Optimality Theory approach where surface forms are stored synchronically, as they reject derivation from a common alternant. Similarly, Marković (2013) proposes that in BCMS, [v] is inserted to avoid hiatus but that assumption only applies to [ov] forms and further rules are necessary for [ev].

Russian verbal morphology studies by Coats (1974); Feinberg (1980); Flier (1974) propose an assimilation rule that simplifies imperfective suffixes, uniting them under /-iv-/ through reduplication, however this approach risks overgeneralisation. Matushansky's (2024: ft. 23) critique identifies a descriptive fallacy here, arguing that [j] in [ov-a]~[u-je] is part of the theme vowel rather than an epenthetic glide, supporting a broader diachronic theory on *yer* vowels relying on identity avoidance (Matushansky 2009; Pesetsky 1979).⁶

Halle (1963) proposes an underlying rounded vowel /o/ for [iv], transforming into [w] or disappearing, but this requires intricate derivational steps. Representational approaches, like that of Melvold (1989), suggest that "/-ova-/ verbs" consist of a sequence of vowels and stress patterns derived through Compensatory Lengthening. However, this model struggles with inter-Slavic variation, as transformations would require either structural differences or reordering of phonological rules. Still, Melvold's (1989) proposal remains one of the most influential studies of Russian morpho-phonology, whose premises are significant later in the present analysis.

Simonović et al. (2023) extend this analysis to BCMS secondary imperfectives by positing that sequences [iv-a]~[u-je] consist of two theme vowels, /i-a/ and /i-e/, with [v] appearing to avoid hiatus. Their tableaux derive optimal candidates that preserve floating features, supporting the view that [u] emerges in the presence of /-je-/ to avoid a consonant cluster. However, although applying this model to the [ov]/[ev] alternations in BCMS is achievable, it predicts that [ev] is the default surface form of the VBLZ in the INF stem and that [ov] is derived via consonant-vowel assimilation with all non-palatals. Such vowel retractions are not in line with BCMS phonotactics, although they are present with velar consonants in Polish and Czech (see Rubach (1984, 2003)).

In sum, synchronic studies, although more concerned with individual languages than the broader picture, clarify that /-ova-/~/-uje-/ involves two separate suffixes whose surface forms are conditioned by phonological context – the VBLZ and the theme vowel. The remaining sections unify diachronic and synchronic perspectives, proposing that all surface forms of the VBLZ stem from one underlying representation, with inter-Slavic variation arising under computational rather than representational differences.

4 Allomorphy is actually phonological derivation

Denominal and deadjectival verbs in Slavic are formed with a suffix that exhibits two types of alternations: (i) intra-linguistically i.e. within the Slavic linguistic family, the suffix alternates between a VC sequence and a single vowel; (ii) inter-linguistically i.e. specific for every Slavic language (group), the VC sequence of the suffix alternates between [ov], [ov] or [ev], and [uv].

The central claim of this paper is that the observed alternations are purely phonological in nature. First, it is evident that the intra-linguistic alternation is not correlated with morpho-syntactic factors such as the type of stem, rather directly dependent on the phonological context. Preceding a vowel-initial suffix like /-a-/, the VBLZ surfaces as a VC sequence, whereas preceding a consonant-initial suffix like the TV /-je-/, the VBLZ suffix emerges as a single vowel. Thus,

⁶Attentive readers will notice that in Matushansky (2024: 6-8) there is no explicit stance on the underlying/inserted status of /j/ in Russian, rather a neutral one, however the idea from Melvold (1989) that the segment itself is a part of the string of morphemes fed into phonology is adopted. The present analysis considers /j/ part of the theme vowel.

the initial part of the alternation must arise due to hiatus resolution, as identified in nearly all diachronic and several synchronic studies before. The second, inter-linguistic aspect of the observed alternation must stem from differences among individual Slavic languages, specifically the linguistic groups they belong to.

The behaviour of this morpheme is specific because hiatuses are rarely resolved in Slavic. Within roots, hiatuses typically surface faithfully, but at morpheme boundaries, the default strategy is vowel deletion (Halle and Matushansky 2006; Olander 2015). Despite this, diachronic and synchronic studies have proposed glide insertion to resolve hiatuses in denominal verbs and secondary imperfectives (see section 3 and, e.g., Rubach 2007). The consonant in the alternation under study is consistently [v], defying co-occurrence predictions, especially for [ev] forms. Thus, linking the [ov]/[ev]/[uv]~[u] alternation to phonotactic glide [v] insertion is untenable. However, there is a debate on the status of [j] in present tense forms—whether it is underlying, derived, or inserted. The question of whether this involves a single theme vowel /-je-/ (Quaglia et al. 2022) or a sequence of /-i-/ and /-e-/ (Matushansky 2024; Melvold 1989) does not significantly impact the present account, as the latter would yield the same result under gliding conditions from /i/ to [j] before /e/ in Melvold (1989). As noted by Casali (1996, 1997), cross-linguistically it is common for a system to exhibit multiple hiatus resolution strategies across different morphological contexts, or even within the same phonological string, depending on the segmental content.

The sole reason for resolving the contexts in focus by alternating between a VC sequence and a single vowel must be a matter of phonological fission (Blevins 2003; Walker 2010, 2011), where an underlying segment "falls apart" in a way to prevent creation of a hiatus, but collapses / fuses into one otherwise (Zaleska 2018). This is achieved by implementing the *Prespecification Method* proposed by Inkelas et al. (1997) and assuming that the source for the fissioned material is an abstract underlying specification. It is expected to have different outcomes for two concrete representations (hiatuses resolved with deletion) and one concrete, one abstract (hiatuses addressed below).

I claim that both the inter- and intra-linguistic aspects of the alternation originate from an abstract UR present in all Slavic languages, who actively adjust the UR to (a) the phonological environment of the construction, and (b) the phonological grammar of the individual language. This study thus shares similar objectives with the diachronic studies discussed in section 3.

An alternation between a sequence and a single segment is not commonly observed cross-linguistically in synchronic terms. However, the Slavic [ov]~[u] alternation bears resemblance to the alternation found in an arbitrarily selected group of verbal bases in the 2nd and 3rd conjugations in Spanish (Boyé and Hofherr 2004; Embick 2012; Harris 1969, 1985): an unstressed mid vowel in the INF is replaced with a diphthong when stressed in the PRS.IND. This alternation is not prevalent in Spanish, as the majority of verbs with a mid vowel and a diphthong do not display the alternation illustrated in (2).

(2) *Spanish diphthong-monophthong alternation*

INF	2SG.PRS.IND	gloss
[sent-'i-r]	['sjent-e-s]	'feel'
[perd-'e-r]	['pjerd-e-s]	'lose'
[dorm-'i-r]	['duerm-e-s]	'sleep'
[poð-'e-r]	['pueð-e-s]	'can'

(Boyé and Hofherr 2004)

For several studies, the arbitrary nature of the alternating verbs in Spanish constitutes compelling evidence in support of stored allomorphs of verbal stems, which are selected to meet phonological requirements (Bermúdez-Otero 2013a; Embick 2012; Linares et al. 2006). However, an earlier perspective put forward by Harris (1969, 1985) contends that the alternation illustrated in (2) stems from an abstract UR. Namely, while the non-alternating segments

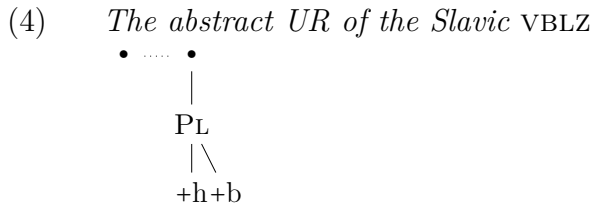
and sequences are inherently well-formed in the underlying structure, the alternating segments/sequences are not inherently well-formed and must be repaired and adapted to the phonological environment upon concatenation. A similar concept is explored in Gussmann’s (2007) analysis of the phonological behaviour of Polish suffixes DIM /-ek/ and POSS /-ow-/, where the tendency for alternation is explained based on the associated/unassociated underlying status of certain segments. The underlying lack of well-formedness is sketched in (3): an alternating diphthong comprises two segmental slots (i.e., X slots or base nodes), but only one of them is specified with segmental content. Due to the prohibition of deletion under stress (cf. *Positional Faithfulness* by Beckman 1998), both slots must be filled. Hiatus avoidance in Spanish leads to the gliding of the first segment, while the second slot is filled with unmarked vocalic content, typically [e] in Spanish. If the alternating diphthong is not stressed, the empty slot remains unrealised and is deleted at the Phonation stage (Trommer 2011).

(3) *Alternating diphthongs in Spanish depending on stress* (Harris 1985: 34)

underlying	surface	
	stressed	unstressed
o	w e	o
X X	X X	X

The Spanish pattern is a mirror image of the Slavic [ov]~[u] alternation, with a slight difference that prosody has no direct influence on the alternation like in Spanish. There is an interesting interaction of the VBLZ morpheme and prosody reported for some Slavic languages (the ones that do not have bound stress like Czech, Macedonian etc.), namely that the VBLZ is *stressophobic* i.e. systematically avoided during stress assignment, cf. Simonović and Mišmaš (2023) for Slovenian, Revithiadou (1999) for Russian*. However, in Slavic, the question of whether the monosegmental ([u]) or the disegmental ([ov]/[ev]/[uv]) realisation will surface is answered based on the segmental content of morphemes attached to the right of the VBLZ.

Therefore, in line with the logic of Harris (1985) for Spanish alternating diphthongs, I propose that the Slavic VBLZ suffix for denominal and deadjectival verbs is underlyingly a defective diphthong with one slot specified, one unspecified (defective in terms of Zimmermann 2017). The autosegmental representations are made explicit in this paper according to (a simplified version of) the Halle-Sagey model of feature organisation *Articulator Theory* (Halle 1992, 1995; Sagey 1986), therefore the VBLZ is underlyingly represented like in (4): it is a sequence of two base nodes (segmental slots), where the first slot is unspecified and the second slot is a high rounded segment, ambiguous between the vowel /u/ and the approximant /v/. The ambiguity is reflected in the Place node (PL) dominating [+high] and [+back], but the [±consonantal] feature with the appropriate value is absent.



Another argument supporting the notion of the unspecified base node in the representation in (4) is derived from loanword integration: /-ov-/ in BCMS and /-uv-/ in Macedonian are employed alongside suffixes like /-a-/, /-ir-/, and /-is-/, so that some bases combine only with /-ov-/ or /-uv-/, some exclusively with /-a-/, /-ir-/, /-is-/, and some with more than one. Analysis of online corpora for both languages reveals that suffixes /-ov-/ and /-uv-/ exclusively combine with consonant-final loaned bases, while there are almost no verbs formed with /-ov-/ and

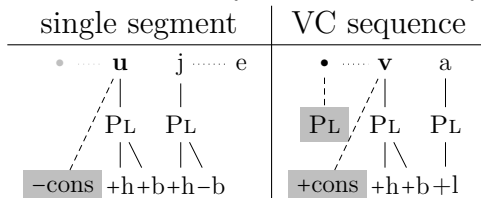
/-uv-/ and a vowel-final base (except the one verb with <5 hits in [Ljubešić and Klubička 2014](#): [rivju-ov-a-ti]/[rivju-u-je-m] ‘review.INF/1SG.PRS’); instead, vowel-final loanwords productively generate verbs with alternative denominal suffixes (cf. BCMS [intervju-is-a-l-a]/[intervju-ir-a-l-a], Macedonian [intervju-ir-a-l-a] ‘interview.L-FORM.F.SG’ or BCMS/Macedonian [parti-j-a-l-a] ‘party.L-FORM.F.SG’). On the other hand, [NKJP \(2010\)](#) lists numerous examples of Polish verbs with a loaned vowel-final stem (5), which leads to a conclusion that there seems to be a mechanism protecting the loaned stem even at the price of hiatus at the left edge of the VBLZ, like *Lexical Conservatism* ([Breiss 2021](#); [Simonović 2015](#); [Steriade 1997](#)) or CRISPEdge ([Itô and Mester 1999](#); [Kaplan 2018](#)). Crucially, hiatus is never found at the right edge of the VBLZ.

(5) *Verbs from loaned vowel-final stems in Polish* ([NKJP 2010](#))

L-FORM.F.SG	2SG.PRS	gloss
[kopi-ov-a-w-a]	[kopi-u-je-ʃ]	‘copy(on device)’
[evaku-ov-a-w-a]	[evaku-u-je-ʃ]	‘evacuate’
[kre-ov-a-w-a]	[kre-u-je-ʃ]	‘create’
[info-ov-a-w-a]	[info-u-je-ʃ]	‘inform’

When forming a denominal verb, the VBLZ suffix is attached to the nominal root, followed by other derivational and inflectional morphemes such as the theme vowel, participle marker, gender/number marker, or person/number marker, depending on the required structure. If the next morpheme begins with a consonant, like the TV /-je-/, the filled segmental slot may be realised as a vowel (supplemented with a [-consonantal] feature), while the vacant slot is deleted, resulting in a single segment. However, if the following morpheme is /-a-/, or any other derivational morpheme commencing with a vowel, the defective diphthong is compelled to disassociate to prevent the formation of an illicit sequence. The specified slot is then enriched with a [+consonantal] feature, yielding [v], while the unspecified slot is developed into a vowel (likely initiated with an epenthetic Place node). Consequently, the underlyingly defective diphthong in (4) never surfaces faithfully due to its inherent non-well-formedness.

(6) *The derivation of Slavic VBLZ surface forms*



The described derivational scenario is closely related to [Harris’s \(1985\)](#) analysis of Spanish, but it is grounded additionally in the diachronic studies on the conditions for monophthongisation of Proto-Slavic diphthongs as discussed by [Andersen \(1972\)](#); [Feldstein \(2003\)](#). The proposed fission of the incomplete diphthong represented in (4) is expected because the two segments in the surface VC sequence have lesser sonority distance on the scale where non-vowels /o/, /e/, and /a/ are most sonorous while high vowels /i/, /u/ and their consonantal counterparts /j/, /v/ are of intermediate sonority. Those sequences are more likely to monophthongise/fuse than sequences with greater sonority distance, more likely to metathetise (like Proto-Slavic /ar/ or /el/) or even lose the consonantal part in the context for monophthongisation (like the /en, on, em, om/ sequences in verbal roots, e.g. BCMS [uzela] / [uzme] ‘take.L-FORM.F.SG/2SG.PRS’).

The derivation in (6) and in the remainder of the paper follows the representational conventions of *Extended Stratal Containment* ([Trommer 2011](#); [Zaleska 2018](#)). Unrealised/un-pronounced material is represented with gray colour, realised material is in black font. Epenthetic material is represented with a gray shading. Underlying association lines between nodes are represented with full lines and epenthetic lines are dashed. Dotted horizontal lines represent precedence re-

lations between members of the same morphological domain, defined as ‘morphological colour’ in van Oostendorp (2006, 2008).

The derivations in (6) only address one part of the alternation, the one that enunciates the difference in phonological environment, yielding the difference between the single segment and the VC sequence. However, as underlined above in Table 3, Slavic languages group based on the quality of the VC sequence. The majority of languages only feature [ov] or [uv], insensitive to the preceding consonant, but a set of Slavic languages shows sensitivity to palatal quality, featuring [ev] as a contextual variant of [ov]. Such a variation can be straightforwardly derived with an Optimality-theoretic grammar (Prince and Smolensky 1993), where variation arises due to differences in the relative ranking of relevant universal constraints. I argue that this is precisely the case in Slavic, namely that the same constraints are active for this alternation, but ranked differently for the three observed groups, which yields the three observed surface shapes.

The UR in (4) is not well-formed because it contains an unspecified base node. A segment which is not specified for a certain feature falls under the auspices of downward well-formedness constraint family SPECIFY, defined like in (7). Since the vowel in (4) is not specified with any features, the constraint is defined simply over F as an abbreviation for height and backness/roundedness.

- (7) • SPECIFY (• → [F])
 ⋮ Assign * for every instance of a segment
 F not specified for features.
(Trommer 2011)

As noticed in many previous studies on this and related alternations, the main power behind the alternation this paper examines is hiatus resolution, that is, avoidance of creating sequences of immediately adjacent vowels. The constraint NOHIATUS is militating against candidates containing adjacent syllable nuclei; it bans candidates with non-underlying hiatuses. Such sequences must be avoided or repaired in order for the constraint in (8) to be satisfied.

- (8) * $\begin{smallmatrix} \sigma & \sigma \\ \mu & \mu \\ \bullet & \bullet \end{smallmatrix}$ NOHIATUS
 Assign * for every sequence R_1, R_2 in I, such that:
 (i) R_1 is dominated by μ_1 , R_2 is dominated by μ_2 ,
 (ii) μ_1 is dominated by σ_1 , μ_2 is dominated by σ_2 .
(Orie and Pulleyblank 2002; Zaleska 2018)

The version of Optimality Theory (Prince and Smolensky 1993) employed here, *Extended Stratal Containment* (Paschen 2018; Trommer 2011; Zaleska 2018; Zimmermann 2017) presupposes that constraints can make reference to the M(orphemic) structure i.e. the input and P(honetic) structure i.e. the output. The constraint DEPENDENCE acts as a ban on some kind of epenthesis (node, feature, association line) by comparing P-structure to M-structure, and MAXIMality prohibits deletion of elements by comparing M-structure to P-structure. Adapted to the needs of the analysis, MAX refers to base nodes, and DEP to features (Lombardi 1998).

- (9) a. MAX Assign * to every instance of a base node (•)
 • present in M, but not in P.
 b. DEP Assign * to every instance of a feature (F)
 F present in P, but not in M.

One of the constraints required for the data in focus is a constraint that prohibits copying of features from one segment to another – *SHARE in (10).⁷ Each of these constraints is violated

⁷*SHARE in (10) is related to constraints such as NOSTRADDLING (Gouskova 2010), UNIQUEAFFILIATION

whenever a candidate shows up exhibiting an illicit change.

- (10) $\begin{array}{c} * \bullet \bullet \\ \diagdown \diagup \\ [-\text{back}] \end{array}$ $\begin{array}{l} * \text{SHARE}[-\text{back}] \\ \text{Assign } * \text{ for every sequence } R_1, R_2, [-\text{back}], \text{ such that:} \\ \text{(i) } R_1 < R_2 \\ \text{(ii) } R_1 \text{ dominates } [-\text{back}] \text{ in M} \\ \text{(iii) } R_1 \text{ and } R_2 \text{ both dominate } [-\text{back}] \text{ in P.} \end{array}$

(Stojković 2023)

The listed constraints ensure that the VBLZ in (4) will surface as a single segment only when there is no danger of creating a hiatus and as a VC sequence when the alternative is a hiatus. The quality of the VC sequence and the differences between three Slavic groups follow from the group-specific hierarchy of the defined constraints. To reduce the number of candidates and shift focus onto the most contrastive examples, only the most probable changes of the input sequences are discussed. Practically, this means that only MAX• (violated by deletion of base nodes) and DEP[+back]/[−high] are taken into account.

The defective diphthong in the exponent of VBLZ in (4) never surfaces faithfully i.e. never surfaces as a sequence of an unspecified base node and a segment which is neither a vowel nor a consonant. This means that SPECIFY must be ranked higher than other constraints. Second, there is no language in which the VBLZ surfaces as a vowel before another vowel – that is a sign of another high ranked constraint, NOHIATUS. Third and final, in order for the deletion of the unspecified node to be a more desirable solution than specifying it with epenthetic material, MAX has to be ranked lower than DEP. The preliminary hierarchy is given in (11).

- (11) *The constraint hierarchy (to be revised)*
SPECIFY(• → [F]); NOHIATUS >> DEP[+back]; DEP[−high] >> MAX•

As mentioned earlier, all inter-Slavic differences in the surface shapes of the VC sequence in the infinitival stem follow from differences in the ranking of constraints universally present in all of their grammars. The following subsections discuss the single vowel [u] form found with the PRS stem, followed by the three VC forms found with the INF stem, respectively. All three VC sequence grammars also derive [u] as the pre-consonantal form.

4.1 The pre-consonantal [u]

In all Slavic languages where the TV /-je-/ follows the VBLZ suffix in the PRS stem, [u] surfaces before /-je-/. As previously mentioned, this result is expected because the alternative [v] would lead to more violations than the optimal candidate, and the fully faithful candidate never emerges. If the alternation were limited to the PRS stem, an additional explanation would be needed for why Macedonian and Bulgarian exhibit [uv] in the present tense. However, in this framework, the answer is simple: the VBLZ is followed by /-a-/ in both the INF and PRS stems in these languages.

Consider (12) as the evaluation of the Lower Sorbian verb [mol-u-je-ʃ] ‘paint, draw’ in 2SG.PRS’. Candidate a. is fully faithful, with no phonological alternations other than concatenation of the VBLZ and the TV /-je-/. This candidate is suboptimal because it contains a node that does not dominate any lower-tier nodes, violating SPECIFY(• → [F]). Candidate c. specifies the empty node with epenthetic features, violating both DEP[+back] and DEP[−high]. The optimal candidate b. simply deletes the problematic node, only violating the lower-ranked MAX•. Thus, candidate c. is harmonically bounded by candidate b. as it incurs more violations, ensuring that b. will always win in this and similar contexts.

(Kawahara 2007b), and GENERALIZED ALIGNMENT (McCarthy and Prince 1986).

(12) *The Slavic pre-consonantal [u]*

Input = a.	$\begin{smallmatrix} * & \sigma & \sigma \\ \mu & \mu & \mu \\ \bullet & \bullet & \bullet \end{smallmatrix}$	$\begin{smallmatrix} \bullet \\ \vdots \\ F \end{smallmatrix}$	DEP [+back]	DEP [-high]	MAX $\bullet$
a. $\begin{smallmatrix} m & o & l & \bullet & j & e \\ & & & P_L & & \\ & & & \backslash & & \\ & & & +h+b & & \end{smallmatrix}$		*!			
b. $\begin{smallmatrix} m & o & l & \bullet & u & j & e \\ & & & P_L & & \\ & & & \backslash & & \\ & & & +h+b & & \end{smallmatrix}$					*
c. $\begin{smallmatrix} m & o & l & o & v & j & e \\ & & & P_L & P_L & & \\ & & & \backslash & & & \\ & & & +b & & & \\ & & & -h & & & \\ & & & +h+b & & & \end{smallmatrix}$			*!	*	

According to this analysis, in every Slavic language, when the VBLZ is succeeded by a consonant-initial morpheme, the surface form of VBLZ will consistently be [u]. Evidence supporting this assertion is drawn from secondary imperfectives in Polish, which are sometimes formed by employing a sequence of two VBLZ suffixes (13).⁸ While the initial occurrence of VBLZ is followed by the vowel-initial /-ow-/, resulting in a sequence, the subsequent occurrence in PRS is followed by a consonant-initial /-je-/, consequently producing [u] as its surface form.

(13) *Secondary imperfectives with /-ow-ow-/ in Polish (NKJP 2010)*

L-FORM.F.SG	2SG.PRS	gloss
[likvid-ov-ov-a-w-a]	[likvid-ov- u -je-ʃ]	‘eliminate’
[obser-ov-ov-a-w-a]	[obser-ov- u -je-ʃ]	‘observe’
[spod-ov-ov-a-w-a]	[spod-ov- u -je-ʃ]	‘cause’
[pʃe-prats-ov-ov-a-w-a]	[pʃe-prats-ov- u -je-ʃ]	‘overwork’

The monophthongal surface form of the VBLZ does not display sensitivity to the preceding consonant in any of the Slavic languages (cf. Table 1–Table 2). There are two principled reasons for this insensitivity: adjacency and specification. First, consonant-vowel interactions in Slavic occur only under immediate adjacency, which is not the case here because the unspecified node intervenes between the root-final palatal and the second node of the VBLZ. Even in candidate (12)-b, where the problematic node is not pronounced (indicated by gray colour of the node), in line with the principles of *Extended Stratal Containment*, it still impedes any potential interaction between a palatal and the second node of the VBLZ. Second, even at later stages of derivation, when a palatal and [u] become adjacent, there is an issue of specification: the second node of the VBLZ in (4) is already specified with backness, and an interaction with a stem-final palatal would require overwriting, i.e., dissociating the underlying [+back] of the VBLZ and sharing [–back] from the palatal. The interaction would also need to have an explicit trigger in form of a markedness constraint on segmental co-occurrence like Rubach’s (2003) PAL, explicitly argued against in Stojković (2023); Zdziebko (2023), who instead argue that palatalisation arises from hidden structure. The remaining sections of the paper demonstrate that the consonant-vowel interaction observed in some Slavic languages (see Table 2) is only possible if the targeted segments are underspecified for backness (and/or if hidden structure in form of floating material is arguably present, cf. Stojković 2023).

4.2 The [ov]-group

Slavic languages from the [ov]-group always feature [ov] as the hiatus-avoiding surface form of the VBLZ in the INF stem, regardless of the preceding consonant’s quality. As a reminder, the group is illustrated with extracted examples from Silesian: (14)-a represents bases ending in

⁸I thank Sławomir Zdziebko for drawing my attention to these data.

non-palatals and (14)-b represents palatal-final roots. In both cases, the VC sequence surfaces as [ov] preceding the vowel-initial suffix /-a-/.

(14) *Silesian denominal imperfectives* (Wencel 2023)

	L-FORM.F.SG	2SG.PRS	gloss
a.	[ɕtop- ov -a-w-a]	[ɕtop- u -je-ʃ]	‘darn’
b.	[pajɕ- ov -a-w-a]	[pajɕ- u -je-ʃ]	‘whip’

In the proposed system, the VC sequence is predicted to only surface when the VBLZ is followed with a vowel-initial morpheme like /-a-/.⁹ The tableau in (15) shows an evaluation of the Silesian verb [ɕtop-**ov**-a-ɕ] ‘darn’, with a base ending in a non-palatal. Like earlier, candidate a. is fully faithful to the input and cannot be optimal because it contains an incomplete base node. Candidate b. leaves the problematic base node unpronounced and candidate c., like in the previous tableau, realises the high back segment as a consonant and fills the unspecified base node as [o] with epenthetic [+back] and [-high]. But here the VBLZ is followed by a vowel-initial suffix, so candidate b. creates a hiatus, which incurs a violation of NOHIATUS. Therefore, candidate c., with epenthetic features and violations of lower ranked DEP constraints, becomes optimal.

(15) *The Slavic pre-vocalic [ov] prevents hiatus*

Input = a.	*σσ μμ ••	! F	DEP [+back]	DEP [-high]	MAX •
a. $\begin{array}{c} \text{ɕ t o p} \bullet \bullet \text{ a} \\ \text{PL} \\ \backslash \\ \text{+h+b} \end{array}$		*!			
b. $\begin{array}{c} \text{ɕ t o p} \bullet \text{ u a} \\ \text{PL} \\ \backslash \\ \text{+h+b} \end{array}$	*!				*
c. $\begin{array}{c} \text{ɕ t o p o v a} \\ \text{PL PL} \\ \text{+b} \backslash \\ \text{-h} \text{+h+b} \end{array}$			*	*	

The consistent emergence of the initial segment as [o] is attributed to the markedness of segmental features. The feature [-high] is cross-linguistically more likely to be unmarked compared to [+high]. Further support for this argument comes from the characteristics of "yer" vowels in Slavic languages, which are commonly realized as mid or low, indicating an association with the [-high] feature. The proposition that [+back] is unmarked as opposed to [-back] has been argued by Rubach (2003); Stojković (2023) based on observations of consonant-vowel interactions in Polish nominal morphology, namely that cases of palatalisation and vowel-to-consonant assimilation are far more frequently found in Slavic as opposed to vowel retractions, seldom present with velar consonants in restricted morpho-syntactic contexts. Consequently, it is reasonable to expect these language-specific patterns to be reflected across the language family.

In this category, the surface form of the VC sequence is insensitive to the quality of the preceding consonant. Even when the base ends in a palatal, languages in this group consistently exhibit the [ov] pattern in verb formation. The lack of evidence for consonant-vowel assimilation, where [-back] might be shared from the consonant to the following vowel, supports the high ranking of *SHARE in the [ov]-group.

This phenomenon is illustrated in (16), showing the failure of consonant-vowel assimilation in the Silesian verb [pajɕ-ov-a-w-a] ‘whip.L-FORM.F.SG’. Key candidates c. and d. both surface with VC sequences: candidate c. features [ev] assimilated to the preceding palatal, while

⁹Another context for [ov] to surface is illustrated with the data in (13), where VBLZ is followed by another instance of VBLZ to create a secondary imperfective.

candidate d. shows [ov] with two epenthetic features, [+back] and [-high]. Given the lack of alternation between [ov] and [ev] in Silesian and other languages belonging to this group, the constraint *SHARE[-back] is ranked above DEP[+back], making the violation in candidate c. fatal. Thus, the optimal candidate d. retains the [ov] pattern even with a preceding palatal.

(16) *The pre-palatal [ov] does not assimilate*

Input = a.	* $\sigma\sigma$ $\mu\mu$ ••	• F	* σ • [-back]	DEP [+back]	DEP [-high]	MAX •
a. p a j t̪ • • a PL \ +h+b		*!				
b. p a j t̪ • u a PL \ +h+b	*!					*
c. p a j t̪ e v a PL PL PL \ +h-b-h +h+b			*!		*	
d. p a j t̪ o v a PL PL \ +b -h +h+b				*	*	

Thus, the VBLZ is not sensitive to the quality of the preceding consonant in this group because the phonological grammar of these languages does not allow sharing/copying of palatality i.e. [-back] and prefers epenthesis of the unmarked feature [+back].¹⁰

(17) *The constraint hierarchy for the /-ov-/-group*

NOHIATUS; SPECIFY(•→[F]) >> *SHARE[-back] >> DEP[-high] >> DEP[+back] >> MAX •

The constraint hierarchy presented in (17) depicts a fundamental aspect of the phonological framework governing the behaviour of the VBLZ suffix and similar morphemes in the linguistic system. Central to this proposal is the imperative to avoid hiatus or the formation of illicit consonant clusters. As a solution, the system generates a vowel preceding a consonant-initial morpheme and a VC sequence preceding a vowel-initial morpheme.

Crucial to the differentiation of this group from other Slavic languages is the specific ranking of constraints in the hierarchy. The constraint requiring specification of an incomplete node takes precedence, provided its implementation does not derive hiatus. This hierarchical arrangement of constraints underlines the preference for epenthesis over sharing at the featural tier, which shapes the distinctive ‘[ov] only’ pattern.

4.3 The [ov]/[ev]-group

Bosnian/Croatian/Montenegrin/Serbian, Russian*, Slovenian and Carpathian Rusyn group together because the VBLZ before /-a-/ in these languages shows sensitivity to the preceding consonant – if preceded by a non-palatal, the VBLZ surfaces as [ov], but if preceded by a palatal, the VBLZ surfaces as [ev]. As a reminder, this is illustrated with Slovenian examples, extracted from Table 1–Table 2. bases ending in a non-palatal are illustrated with (18)-a, while (18)-b

¹⁰An anonymous reviewer expresses concerns that the high-ranked ban on sharing [-back] in (16) incorrectly predicts lack of palatalisation in [ov]-only languages. However, following Czaplicki (2013); Gussmann (2007); Stojković (2023); Zdziebko (2023) rather than Padgett (2010), palatalisation is an effect of hidden structure, which is not affected by *SHARE[-back] as defined in (10), to ban shared associated material. In line with Trommer’s (2011) approach, such grammatical repairs occur up to the Phonation stage, with subsequent phonetic adjustments like Surface Palatalisation being governed by strict adjacency and articulatory factors.

illustrates palatal-final stems. The [ov]~[ev] alternation, additional to [ov]~[u], is only observed in the context where the VBLZ suffix is followed by /-a-/. When followed by the theme vowel /-je-/, the VBLZ still surfaces as [u].

(18) *Slovenian denominal imperfectives* (Herrity 2000)

	L-FORM.F.SG	2SG.PRS	gloss
a.	[pot- ov -a-l-a]	[pot- u -je-ŋ]	‘travel’
b.	[deʒ- ev -a-l-a]	[deʒ- u -je-ŋ]	‘rain’

Note that regardless of the [ov]~[ev] alternation, the VBLZ is always realised as [u] before the TV /-je-/. The consistent realisation of the VBLZ suffix as [u] instead of, say, [i], in the presence of /-je-/ in this group serves as evidence that the [-back] feature essential for the formation of [ev] is not underlyingly present for the VBLZ, rather acquired elsewhere (via epenthesis or sharing). This is an additional argument that the distinctions observed among the three groups stem from differences in the phonological grammar rather than representational differences in the underlying structure of the VBLZ.

For the surface form [ov] the analysis is not much different from the one presented in (15). The tableau in (19) shows an evaluation of the Slovenian verb [pot-ov-a-l-a] ‘travel.L-FORM.F.SG’, where the base ends in a non-palatal. Candidate a. represents the fully faithful candidate, which is rejected by SPECIFY(•→[F]) due to containing an unspecified base node. Candidate b. features [u] as the surface realisation of VBLZ, which is suboptimal because the following segment is also a vowel. Two adjacent vowels create a hiatus, disfavoured by the undominated NOHIATUS. The optimal candidate c. fills the unspecified node with epenthetic material, violating lower-ranked DEP constraints.

(19) *The pre-vocalic [ov] prevents hiatus*

Input = a.	$\begin{smallmatrix} * \sigma \sigma \\ \dots \end{smallmatrix}$ $\begin{smallmatrix} \text{u} \text{u} \\ \dots \end{smallmatrix}$	$\begin{smallmatrix} \bullet \\ \vdots \\ \text{F} \end{smallmatrix}$	DEP [+back]	DEP [-high]	MAX •
a. $\begin{smallmatrix} \text{p o t} \bullet \bullet \text{ a} \\ \text{PL} \\ \backslash \\ \text{+h+b} \end{smallmatrix}$		*!			
b. $\begin{smallmatrix} \text{p o t} \bullet \text{ u a} \\ \text{PL} \\ \backslash \\ \text{+h+b} \end{smallmatrix}$	*!				*
c. $\begin{smallmatrix} \text{p o t o v a} \\ \text{+b} \text{ PL PL} \\ \text{-h} \text{ } \backslash \\ \text{+h+b} \end{smallmatrix}$			*	*	

In this group, the surface form of the VBLZ shows sensitivity to the preceding consonant. When the bases end in a palatal consonant, this group consistently exhibits [ev] followed by /-a-/, and [u] followed by /-je-/. Evidence supporting consonant-vowel assimilation as a process of sharing the feature [-back] from the consonant to the following vowel as an argument for ranking *SHARE lower in the [ov]/[ev]-group. The phonological grammar prefers sharing [-back] over epenthesis of [+back].

The tableau in (20) exemplifies assimilation with the evaluation of the Slovenian verb [deʒ-ev-a-l-a] ‘rain.L-FORM.F.SG’. The fully faithful candidate a. and candidate b. violate highest-ranked constraints and are therefore suboptimal. Candidates c. and d. avoid the hiatus by realising the right node as a consonant and epenthesising the unmarked [-high] (as opposed to the marked [+high]). The only difference is in the origin of backness. In c. [-back] is shared from the palatal to the vowel and in d. [+back] is epenthesised. DEP[+back] outranking *SHARE[-back] derives the preference for sharing over epenthesis, realised as the distinctive [ov]/[ev] pattern dependent on the presence of a palatal.

(20) *The post-palatal [ov] assimilates to [ev]*

Input = a.	$\begin{smallmatrix} * \sigma \sigma \\ \text{PL} \end{smallmatrix}$	$\begin{smallmatrix} \bullet \\ \vdots \\ F \end{smallmatrix}$	DEP [+back]	DEP [-high]	$\begin{smallmatrix} * \bullet \bullet \\ \backslash / \\ [-\text{back}] \end{smallmatrix}$	MAX $\bullet$
a. $\begin{smallmatrix} d e \bar{3} \bullet \bullet a \\ \text{PL} \\ \backslash \\ +h+b \end{smallmatrix}$		*!				
b. $\begin{smallmatrix} d e \bar{3} \bullet u a \\ \text{PL} \\ \backslash \\ +h+b \end{smallmatrix}$	*!					*
c. $\begin{smallmatrix} d e \bar{3} e v a \\ \text{PL PL PL} \\ / \backslash / \backslash \\ +h-b-h-h+b \end{smallmatrix}$				*	*	
d. $\begin{smallmatrix} d e \bar{3} o v a \\ \text{PL PL} \\ / \backslash \\ +b-h-h+b \end{smallmatrix}$			*!	*		

The constraint hierarchy in (21) reflects the pattern established for the [ov]/[ev]-group: avoiding hiatus requires the abstract VBLZ to surface as a VC sequence, a consonant at the right edge and a vowel at the left. The vowel is [-high] by default, with a preference for frontness if [-back] can be acquired via sharing, but if not, [+back] will be epenthesised.

(21) *The constraint hierarchy for the [ov]/[ev]-group*

NOHIATUS; SPECIFY($\bullet \rightarrow [F]$) $\gg$ DEP[+back] $\gg$ DEP[-high] $\gg$ *SHARE[-back] $\gg$ MAX $\bullet$.

All palatals in the languages in this group participate in the [ov]~[ev] alternation, however, not all bases. Namely, the VBLZ suffix alternating between [ov]~[ev] is also productive for the derivation of verbs from loaned bases in BCMS, Russian*, and Carpathian Rusyn (in Slovenian the suffix /-ir-/ is employed for these purposes). Loaned palatal-final bases fall out of the established [ov]~[ev] pattern in the sense that they only combine with [ov] in BCMS¹¹ and Carpathian Rusyn, while in Russian* they alternate [ov]/[ev]. (22) lists some examples recovered for the languages in the [ov]/[ev] group.¹²

(22) *Verbs from loaned bases in the [ov]/[ev]-group*

Bos/Cro/Mon/Ser	[lintʃ-ov-a-ti]	‘lynch’
	[paʃ-ov-a-ti]	‘rule in a ruthless manner’
Carpathian Rusyn	[ʃverts-ov-a-ts]	‘smuggle’
	[angaʒ-ov-a-ts]	‘engage’
Russian*	[keʂ-<ev>-a-t’]	‘cache’
	[lintɕ-<ev>-a-t’]	‘lynch’

¹¹Suffix /-ov-/ combines with loaned bases in Serbian, Montenegrin and (some) Bosnian varieties, in addition to /-ir-/ and /-is-/, while Croatian varieties almost exclusively use /-ir-/ for older (Latinate) loanwords, and /-a-/ for newer (i). See Simonović (2015); Simonović and Samardžić (2013) for an account on this matter.

(i)	Croatian	Bos, Mon, Ser	gloss
	[kontrol-ir-a-l-a]	[kontrol-is-a-l-a]	‘control’
	[angaʒ-ir-a-l-a]	[angaʒ-ov-a-l-a]	‘engage, hire’
	[lajk-a-l-a]	[lajk-ov-a-l-a]	‘like on social networks’
	[gugl-a-l-a]	[gugl-a-l-a]	‘google, search online’

¹²For Carpathian Rusyn, the examples are collected from the cited sources. Examples for BCMS are collected from Simonović (2015) and checked with the web-based corpora srWaC (Ljubešić and Klubička 2014) and MaCoCu (Bañón et al. 2022). For Russian, examples are collected from two corpora: ruTenTen (Jakubíček et al. 2013; Suchomel and Pomikálek 2012) and JSI Timestamped Russian corpus (Bušta et al. 2017; Trampus and Novak 2012). The attentive reader should bear in mind that checking the Russian examples on social networks also showed sporadic uses of [ov] with loaned roots, indicating a potential change in progress.

Simonović (2015) ascribes the observed resistance of loaned palatals to *Lexical Conservatism* (Breiss 2021; Steriade 1997), the principle that protects loanwords because it assumes that they enter the mental lexicon as surface forms, which makes them representationally different from native forms. A way to implement this assumption in the system employed here is to assume lexicon stratification (Itô and Mester 2001; Itô and Mester 1999), where the loanwords are integrated and stored as stems, in contrast to the native lexicon, comprised of roots.¹³ Bases are then evaluated at the stem stratum, while stems are evaluated at the word stratum, together with the suffixes concatenated at the respective stratum (Bermúdez-Otero 2013b). Instead of selecting an allomorph, the grammar actually needs to address the ranking of *SHARE[–back] at the word stratum. For BCMS and Carpathian Rusyn, it would be necessary to rank the ban on sharing above DEP constraints. When faced with the candidates like in (20), the grammar in (23)-a would derive [ov] even before a palatal.

- (23) *Word stratum grammar for BCMS and Carpathian Rusyn*
 NOHIATUS; SPECIFY(•→[F]) >> *SHARE[–back] >> DEP[+back] >>
 DEP[–high] >> MAX •

For Russian*, the loanwords, presumably either stored as bases or stored as stems, but evaluated without this particular reranking, follow the established pattern of [ov]~[ev] alternation, predicted by the hierarchy in (21).¹⁴

The analysis presented here relies heavily on two empirically grounded assumptions. First, it relies on the assumption that only [–back] is active in Slavic assimilation processes i.e. can be shared. Namely, consonant-vowel alternations in Slavic are generally based on the influence of [–back] from one segment to another, reflected in the plethora of palatalisation patterns (with very rare cases of [+back] activity, like the Polish vowel retraction in the LOC.M/N.SG or Czech backness assimilation in VOC.M/N.SG, both arguably arising from floating backness). Second, the analysis relies on the assumption that only certain consonants may interact with vowels, but only in a certain manner. Palatals regularly influence the quality of vowels in Slavic, but other consonants almost never interact, primarily coronals and labials, which can in turn be influenced by vowels (i.e. in palatalisation). In Stojković (2023) the difference in activity of [+back] and [–back] is grounded in markedness, in line with de Lacy (2006); Inkelas et al. (1997): marked elements are more likely listed (provided by the lexicon), active in assimilation and overwriting processes through *Preservation of the Marked*, while unmarked elements are more likely inserted by the grammar and frequently lost in the derivation. Thus, the [ov]~[ev] pattern gives another argument in favour of Stojković (2023) but also Rubach (2003) that [+back] is unmarked in Slavic. The property of certain consonants that they are prone to influence from the adjacent vowels is ascribed to structural deficiency, namely, underlying underspecification for backness, which is then either shared from the vowel ([–back], leading to palatalisation) or epenthesis ([+back], non-palatal).

Finally, the argument that the empty slot is filled with [–high] is logical for the majority of Slavic languages given the surface forms of *yer* vowels, but it might seem problematic for BCMS,

¹³A similar proposal was put forward for the treatment of epenthesis in BCMS, where loanwords are treated as stems, evaluated at the word stratum, which explains the optionality of [a] epenthesis with both licit and illicit consonant clusters in loanwords (Stojković 2021, but see also Bermúdez-Otero 2019 for similar arguments from Spanish).

¹⁴An anonymous reviewer suggests an alternative interpretation of the data in (22), where palatals in loanwords would have different URs for speakers of Slavic languages: as an underlyingly non-palatalised segment (Carpathian Rusyn and BCMS) or as an underlyingly palatalised one (Russian). This interpretation aligns closely with the present proposal, as both build upon a representational difference. A potential setback for Carpathian Rusyn and BCMS is that no perceptual differences are reported for underlying, derived, and loaned palatals; in fact, Wierzchowska (1971) provides strong evidence that this holds in Polish as well. However, see Botinis et al. (2011) for a report on durational differences between underlying and derived palatals in Greek.

where [a] is arguably the default vowel (Simonović 2015; Zec 2002), employed for breaking illicit clusters, particularly at the right edge of the word.

- (24) *Vowel epenthesis in BCMS*
- | | | |
|------------|--------------|--------------------|
| [jesam] | [jesmo] | ‘be.1SG~1PL’ |
| [koverat] | [koverta] | ‘envelope.NOM~GEN’ |
| /iz-brati/ | [iz-a-brati] | ‘choose.INF’ |

In an Optimality-theoretic grammar, such an outcome, that one vowel quality is employed when a full vowel is inserted and another when a single feature is needed, is modelled as a hierarchy of constraints like the one in (25) essentially says that [a] will surface in the context of an illicit cluster, but not when an unspecified node needs to be specified. In other words, the epenthesis of a base node requires [+low]. Bear in mind that [+low] and [−high] are still compatible when it comes to creating an [a], but the former is only inserted together with a base node.

- (25) *The partial hierarchy of epenthesis in BCMS*
 *CC >> DEP[+low] >> SPECIFY >> DEP[−high]

Thus, BCMS is relatively unique in having different default epenthetic vowel [a] and default epenthetic feature(s) [−high]. Such grammars are not uncommon cross-linguistically (Kawahara 2007a; Rice 2007). Other Slavic languages arguably do not exhibit this difference, as their successors of *yer* vowels are consistently mid.

4.4 The [uv]-group

Bulgarian, Lemko Rusyn, Macedonian, and Ukrainian imperfective verbs derived from nominal and adjectival bases surface with [uv] where [ov] or [ov]/[ev] is found in other Slavic languages. This special fissioned shape of the VBLZ is, similarly to [ov], not sensitive to the quality of the previous consonant. The default form of the VBLZ is illustrated in (26)-a, with the base ending in a non-palatal, while the lack of interaction with palatals in the [uv]-group is exemplified in (26)-b, both with verbs from Lemko Rusyn.

- (26) *Lemko Rusyn imperfectives* (Fontański and Chomiak 2000)
- | | L-FORM.F.SG | 2SG.PRS | gloss |
|----|--------------------------|------------------------|---------|
| a. | [ris- uv -a-l-a] | [ris- u -je-ʃ] | ‘draw’ |
| b. | [oboʒ- uv -a-l-a] | [oboʒ- u -je-ʃ] | ‘adore’ |

At first glance, the reason for this kind of behaviour from the perspective of the present analysis might be that epenthesis of both [+back] and [−high] is prohibited, and that phonological grammars of these languages prefer sharing at the Place node tier. There is however no empirical grounding for such an interpretation. Place node sharing is typical for segmental harmony, which is untenable in the case of the [uv]-group because none of these languages feature full regressive assimilation from consonant to the vowel or vowel to vowel elsewhere in their phonology.

A more plausible scenario is that these languages are subject to a different markedness hierarchy, which lists [+high] as the unmarked value of [±high] (in terms of Dresher (2009)). It follows that [+high] is preferably epenthetic, and [−high] is preferably listed. Such a situation is yielded by a system where epenthesis of [−high] is prohibited i.e. DEP[−high] outranks *SHARE[−back] and DEP[+back]. The surface realisation [uv], where the left node of VBLZ is realised as a high back vowel, thus arises from a ban on [−high] epenthesis. When the VBLZ is found following a non-palatal, the system has no other option but to epenthesise [+back] in order to satisfy SPECIFY and produce a well-formed segment.

The tableau in (27) shows an evaluation of the Lemko Rusyn verb [ris-uv-a-ti] ‘draw’. The

fully faithful candidate is suboptimal due to the unspecified base node it contains, a violation of the undominated SPECIFY($\bullet \rightarrow [F]$). With the candidate b. violating NOHIATUS and thus being excluded from further consideration, the question remains whether the surface representation will contain [o] or [u]. The decision is passed on to DEP[-high], which rejects candidate c. for epenthesising [-high] and favours candidate d. with an epenthetic [+high] as the optimal candidate.

(27)

Input = a.	$\begin{smallmatrix} * \sigma \sigma \\ \mu \mu \end{smallmatrix}$	$\begin{smallmatrix} \bullet \\ ! \\ F \end{smallmatrix}$	DEP [-high]	$\begin{smallmatrix} * \bullet \bullet \\ \backslash / \end{smallmatrix}$ [-back]	DEP [+back]	MAX $\bullet$
a. $\begin{smallmatrix} r & i & s & \bullet & \bullet & a \\ & & & P_L & & \\ & & & \backslash & & \\ & & & +h+b & & \end{smallmatrix}$		*!				
b. $\begin{smallmatrix} r & i & s & \bullet & u & a \\ & & & P_L & & \\ & & & \backslash & & \\ & & & +h+b & & \end{smallmatrix}$	*!					*
c. $\begin{smallmatrix} r & i & s & o & v & a \\ & & & P_L & P_L & \\ & & & / \backslash & / \backslash & \\ & & & +b-h & +h+b & \end{smallmatrix}$			*!		*	
d. $\begin{smallmatrix} r & i & s & u & v & a \\ & & & P_L & P_L & \\ & & & / \backslash & / \backslash & \\ & & & +b+h & +h+b & \end{smallmatrix}$					*	

The evaluation in (28) answers the question why [uv] is not sensitive to the quality of the preceding palatal. Like in the previous tableaux, the fully faithful candidate a. is rejected by SPECIFY($\bullet \rightarrow [F]$). Candidates b. and c., with mid vowels [o] and [e] as realisations of the left base node of the VBLZ, are both excluded by DEP[-high] for suggesting an epenthetic [-high], regardless of whether they acquire backness via epenthesis of [+back] or via sharing of [-back] from the palatal. Candidates d. and e. both epenthesise [+high], which avoids the violation of DEP[-high] and creates space for both [i] and [u] to arise post-palatal. Candidate d. with [uv] wins over candidate e. with [iv] because the latter violates *SHARE[-back].

(28)

Input = a.	$\begin{smallmatrix} * \sigma \sigma \\ \mu \mu \end{smallmatrix}$	$\begin{smallmatrix} \bullet \\ ! \\ F \end{smallmatrix}$	DEP [-high]	$\begin{smallmatrix} * \bullet \bullet \\ \backslash / \end{smallmatrix}$ [-back]	DEP [+back]	MAX $\bullet$
a. $\begin{smallmatrix} o & b & o & \text{3} & \bullet & \bullet & a \\ & & & & P_L & & \\ & & & & \backslash & & \\ & & & & +h+b & & \end{smallmatrix}$		*!				
b. $\begin{smallmatrix} o & b & o & \text{3} & o & v & a \\ & & & & P_L & P_L & \\ & & & & / \backslash & / \backslash & \\ & & & & +b-h & +h+b & \end{smallmatrix}$			*!		*	
c. $\begin{smallmatrix} o & b & o & \text{3} & e & v & a \\ & & & & P_L & P_L & P_L \\ & & & & / \backslash & / \backslash & / \backslash & \\ & & & & +h-b & -h & +h+b & \end{smallmatrix}$			*!	*		
d. $\begin{smallmatrix} o & b & o & \text{3} & u & v & a \\ & & & & P_L & P_L & \\ & & & & / \backslash & / \backslash & \\ & & & & +b+h & +h+b & \end{smallmatrix}$					*	
e. $\begin{smallmatrix} o & b & o & \text{3} & i & v & a \\ & & & & P_L & P_L & P_L \\ & & & & / \backslash & / \backslash & / \backslash & \\ & & & & +h-b & +h & +h+b & \end{smallmatrix}$				*!	*	

The third constraint ranking, given in (29), depicts the system where [uv] will always win over [ov] and [ev] in the context of /-a-/, because hiatus must be avoided and all nodes must be specified, but [-high] should not be epenthetic. Instead, the languages of the [uv]-group epenthesise [+high], but still remain insensitive to the quality of the preceding consonant.

- (29) *The constraint hierarchy for the [uv]-group*
 NOHIATUS; SPECIFY($\bullet \rightarrow [F]$) $\gg$ DEP[-high] $\gg$ *SHARE[-back] $\gg$ DEP[+back] $\gg$ MAX $\bullet$.

Bear in mind that all identity between the derived vowel and the consonant of the VBLZ is thus accidental in this system. The vowel never acquires features from the right, always from the left (a feature which Slavic languages have developed, not inherently preserved since Proto-Slavic had both progressive and regressive palatalisation, but modern-day Slavic only feature regressive), which can be ascribed either to high-ranked CRISPEGE (Itô and Mester 1999; Kaplan 2018) or to a general ban on leftward sharing (Finley 2008). Thus for Lemko Rusyn and Ukrainian, the abstract underlying representation (4) gives rise to [u] when followed by /-je-/, in line with the derivation in (15), while the same grammar, supplemented in (29), derives the sequence [uv] preceding /-a-/, not sensitive to the preceding consonant.¹⁵

Finally, it is assumed here that Macedonian and Bulgarian also have access to the proposed abstract representation of the VBLZ morpheme (4), derived as [uv] in the same way like in Lemko Rusyn and Ukrainian. The fact is that these languages have completely lost the INF/PRS stem difference for the verbs in focus. The VBLZ is in these languages always followed by /-a-/. From the perspective of the present analysis, the vocalic context provided by /-a-/ results in the possibility to realise the VBLZ in (4) as a sequence. In other words, if Bulgarian and Macedonian also feature the abstract representation of the VBLZ, the fact that it is always followed by a vowel regardless of the stem type will render only [uv] as the surface representation, parallel to the evaluations given in (27)–(28). The denominal and deadjectival VBLZ verbs are therefore never met with the phonological environment for simplification of the abstract sequence into the single segment [u] like in (12).

If the explanation above is accepted, the question remains why is the phonological environment for the derivation of [u] absent from Bulgarian and Macedonian. As a reminder, denominal and deadjectival verbs in these languages never show an alternation for the VBLZ between infinitival and present tense stem (the same is true for secondary imperfectives). Firstly, some kind of morphological "transfer" of the INF stem into the PRS is out of the question because the difference between the two stems is well preserved for other types of verbs. However, there must be something in the morphological development of the VBLZ suffix in these languages that requires combining with /-a-/ exclusively, or diachronic changes in the distribution of /-je-/ as a theme vowel have taken place. The examples in (30) show that the /-je-/ is available as a PRS theme vowel in Bulgarian and Macedonian, only not for the [uv]-verbs.

(30)		2SG.PRS	2PL.PST	gloss
Bulgarian		[kaʒ-e-j]	[kaz-a-x-te]	‘say’
		[sutʃ-e-j]	[suk-a-x-te]	‘knead’
		[tsel-uv-a-j]	[tsel-uv-a-x-te]	‘kiss’
Macedonian		[laʒ-e-j]	[lag-a-v-te]	‘lie’
		[platʃ-e-j]	[plak-a-v-te]	‘cry’
		[ver-uv-a-j]	[ver-uv-a-v-te]	‘believe’

¹⁵An anonymous reviewer proposes an alternative approach, namely to consider that a speaker of Lemko Rusyn and Ukrainian operates with a different UR, the sequence of a high back rounded vowel [u] followed by an underspecified glide that is realized as [j] before the front vowel /-e-/ and as [w] (surface [v]) before a back one /-a-/. While interesting at the level of individual languages, such an approach is contrary to the main goal of the paper, inspired by a growing body of phonological literature (Bermúdez-Otero 2012; Bye and Svenonius 2012; Faust et al. 2018; Lahrouchi and Ulfsbjorninn 2024; Lampitelli and Ulfsbjorninn 2023; Parfene and Ulfsbjorninn 2023; Paschen 2018; Scheer 2016; Simonović et al. 2023; Stojković 2023; Stojković and Simonović 2026; Trommer 2020; Trommer and Zimmermann 2014; Zimmermann 2017): to explore the possibility of underlying structural similarity of surface realisations with their differences arising from computational preferences. The fact that the explored languages are genetically related is seen as an additional argument in favour of such exploration.

Koneski (1986) relates this matter to the desolation of the /e/-class, where verbs migrated either into the /i/-class or /a/-class in the scriptures from the 18th century, presumably due to morphological analogy (1986: 188). Earlier written sources of Macedonian already from the 13th century show the alternation [uv]~[u] like in contemporary Ukrainian, in parallel to the scarcely present [uv]-only forms (1986: 168-170). Even much more contemporary sources, like the newspaper *Pionerski vesnik* from 1945, feature more verbs with [u-e] than with [uv-a] in the present tense, proportionally 8 : 2 (Arno.mk 2026). Markovikj (2001) reports that the [uv]~[u] alternation parallel to Ukrainian and Lemko Rusyn is still present in the northeastern dialects of Macedonian (2001: 174-175), while the westernmost dialects feature non-alternating [u-a] and [o-a] (2001: 51), meaning that the ban on hiatuses is arguably ranked lower in those than in other Slavic systems.

An interpretation similar to Koneski's is given by Ganeva (2018) – the change is ascribed to analogy, but the transfer to *a*-class is attributed to imperfectivisation, with the formant /-a-/ as the main culprit. In other words, the rise of /-a-/ as the marker for [-perfective] has contributed to the disappearance of the context for phonological alternation triggered by the hiatus (Ganeva 2012: 16). Interestingly enough, Haralampiev (1983); Ivanova Mircheva and Haralampiev (1999) observe the correlation between the [uv]~[u] alternation and stress placement in western Bulgarian dialects, where the VBLZ was unstressed and followed by /-e-/ in the present tense (and stressed later again, cf. [gospoḡ-u-je-m] 'act like a lady'), while eastern and central dialects maintain a stressed [uv] followed by /-a-/. Finally, Radeva (1979) lists a plethora of denominal verbs in Bulgarian dialects, including verbs with non-alternating [ov-a] like [panagir-**ov**-a-m] 'go to a fun fair', characteristic for the southeastern, Rup(ian) dialects.

Thus, class desolation processes (rise of IPFV /-a-/ and stress shift) have contributed to the presumed "stabilisation" of the suffix from the proposed abstract representation into the VC sequence, which consequently never alternates. From the perspective of learnability, speakers of Macedonian and Bulgarian could be split into minimally two groups: one group (let us call them speakers of standards) only ever hears and learns denominal verbs with the sequence [uv] between the base and /-a-/. It could be that these speakers today only have the VC sequence as the exponent for VBLZ. On the other hand, given *Richness of the Base* (Prince and Smolensky 1993), there are no theoretical obstacles to the claim that the abstract representation is present in the mental lexicon even though its surface realisation does not reveal it.

The other group of speakers of Bulgarian and Macedonian are speakers of non-standard dialects, which have to operate with multiple surface representations in similar contexts (alternating [uv]~[u] vs. non-alternating [uv], non-alternating [ov] vs. [uv], [o-a] vs. [uv-a], and [u-a] vs. [uv-a]). Following Pertsova (2007); Prince and Tesar (2004); Tesar and Smolensky (2000), I assume that speakers confronted with multiple realisation of the same morpheme are able to make hypotheses about abstracts representations since the alternations do not really follow from the acquired surface phonotactics, as evidenced by McQueen et al. (2006). The non-standard speakers of Bulgarian and Macedonian thus encounter verbs with, say, [ov] and [uv] and, without phonotactic information that can differentiate their usage, make assumptions about the UR and different grammars depending on the register (Guy 1997; Kroch 1989).

The proposed analysis however covers both scenarios. The last generation of L1 speakers of standard dialects that can be pin-pointed in the development of Bulgarian and Macedonian arguably actively used both [u] and [uv] as surface realisations of the VBLZ, for which there was arguably a phonological context for the formation of [u] before /-je-/. Subsequent generations, for which diachronically confirmed morphological changes have removed the context for the phonological alternation that would have actively taken place, are arguably learning and storing the sequence [uv] or actively deriving alternative dialectal and standard forms before /-a-/ (in line with the findings of studies on artificial learning).

4.5 Discussion

Slavic languages constitute three groups based on the shape of the VC sequence: the [ov]-group, the [ov]/[ev] group, and the [uv] group. This is shown to arise from computational differences, modelled as three OT grammars in (17), (21), and (29). Assuming that NOHIATUS and SPECIFY($\bullet \rightarrow [F]$) are undominated, the relative rankings of the three constraints give the factorial typology in (31). The most languages exhibit the [ov] pattern, followed by [ov]/[ev], while [uv] is the least likely to arise. Note that in the [ov]/[ev] pattern, [ev] is seen as the less costly option, while [ov] arises because sharing of [+back] is not an available option for representational reasons.

- (31) *Factorial typology derives the observed three groups*
- a. *SHARE[−back] >> DEP[+back] >> DEP[−high] $\mapsto$ [ov]
 - b. *SHARE[−back] >> DEP[−high] >> DEP[+back] $\mapsto$ [ov]
 - c. DEP[+back] >> *SHARE[−back] >> DEP[−high] $\mapsto$ [ov]/[ev]
 - d. DEP[+back] >> DEP[−high] >> *SHARE[−back] $\mapsto$ [ov]/[ev]
 - e. DEP[−high] >> *SHARE[−back] >> DEP[+back] $\mapsto$ [uv]
 - f. DEP[−high] >> DEP[+back] >> *SHARE[−back] $\mapsto$ [uv]/[iv] ?

Factorial typology of the three constraints on sharing and epenthesis derives the observed three patterns, with the addition of [uv]~[iv], the pattern so far unattested in Slavic. In an ideal world, there would be a Slavic system where [uv] is sensitive to the quality of the preceding consonant. Although much more research is needed, it might be fruitful to conduct an extensive search for such examples in the future, since pairs of secondary imperfectives like (32) are scarcely present in the Ukrainian corpora ParlaMint-UA (Kopp et al. 2023), while the available dictionaries only list the [uv] version. However, no denominal or deadjectival verbs with the same optionality have been identified yet so the data should not be considered conclusive.

- (32) *The Ukrainian [uv]~[iv] pairs*
- | | | |
|------------------|------------------|----------------|
| [oplatʃ-uv-a-ti] | [oplatʃ-iv-a-ti] | ‘pay’ |
| [ozvuʃ-uv-a-ti] | [ozvuʃ-iv-a-ti] | ‘record sound’ |

The fact that such a Slavic language is not reported (to the best of the author’s knowledge) may be a systematic gap, too, and the literature on markedness effects speaks in favour of such interpretation. Namely, the factorial typology in (31) shows that theoretically there are no obvious obstacles for a Slavic language with an [uv]~[iv] pattern. However, studies of palatalisation and vowel retraction in Polish suggest that vowel frontness is more marked than backness (Gussmann 2007; Rubach 2003; Stojković 2023; Zdziebko 2023) and a system like (31)-f is the mirror image of those claims. Namely, in a phonology based on DEP[+back] outranking *SHARE[−back] like (31)-f, [+back] would be the marked value of this feature, which would have severe consequences for the entire system, including predicting failure of palatalisation. Given that palatalisation is one of the defining features of Slavic phonology, dependent on [−back] being marked, it could be that the language following the grammar in (31)-f is not Slavic to begin with. The presence of Rusyn dialects in 3/4 predicted groups empirically supports the impression that an [uv]~[iv] pattern is excluded in Slavic for independent, markedness-related reasons.

Other arguments, related to the markedness of [−back] in Slavic, have been made more recently that /i/ falls under the auspices of *Comparative Markedness* (McCarthy 2003) in BCMS, meaning that there is an active mechanism in the phonology which resist having a derived /i/ in the output, as opposed to an underlying /i/, which is obligatorily preserved even as iotation of the preceding consonant (Simonović et al. 2023; Stojković and Simonović 2026). The extent to which this particular solution can be applied to the [uv]-group is yet to be examined, given the productivity of alternations involving /i/ in Ukrainian (Foster 1966; Rubach 2005).

5 Conclusion

This paper sheds light onto an alternation widespread in Slavic languages: the alternation of the VBLZ suffix between a VC sequence in the INF and a single vowel in the PRS stem. It is first shown that the alternation is not morphologically, rather phonologically conditioned – the sequence only appears preceding a vowel and the single vowel appears preceding a consonant.

Slavic languages thus exhibit the [u]~[ov]/[ev]/[uv] alternation when deriving imperfective verbs, dependant on the phonological environment, and constitute three groups on the basis of the surface quality of the VC sequence: the [ov]-group, the [ov]/[ev] group, and the [uv] group.

By recourse to the diachronic studies on the development of the VBLZ from PIE **h₂eu_h₁-* and employing the *Prespecification Method* (Inkelas et al. 1997), the VC~V (i.e. [ov]~[u]) alternation is shown to arise from an abstract UR: the VBLZ exponent is a diphthong which never surfaces faithfully, rather ‘fuses’ into [u] before /-je-/ to avoid a cluster (or rather simplifies into a single vowel, in terms of Zaleska 2018) and ‘fissions’ into a VC before /-a-/ to avoid a hiatus. The decision between fission and fusion is given over to the general requirements for string well-formedness, primarily hiatus resolution, diachronically confirmed by numerous studies. Thus the complexity of the pattern is reflected in the representation, while the proposed grammars are awarded simplicity of general phonological properties, found outside of narrow morpho-syntactic contexts. The simplicity of the proposal allows straightforward extension to other morphemes with similar phonological behaviour, like the secondary imperfectives with [iv]~[u] and semelfactives with [n]~[∅].

The proposed abstract UR, inspired by Harris (1985) for Spanish, is unique for all Slavic languages and all arguably have access to its content. While the principal part of the pattern is ascribed to the representational particularities of the morpheme in question, the three groups are shown to arise from computational particularities of the phonological grammar, reliant on two autosegmental principles: feature epenthesis and feature sharing. The factorial typology of the relevant lower-ranked constraints straightforwardly derives all three observed patterns, but also gives insight in the directions for future research.

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